



ELEMENTARY MATH (CALCULUS)



DEFINITE INTEGRALS: AREA BETWEEN CURVES



LEARNING OBJECTIVES

Completion Upon of this unit, you should be able to:

- Obtain the area bounded by a given function and the axis of definition.
- Obtain the area bounded by any two given functions.
- Obtain the area bounded by any two given functions intersecting at a point.





INTRODUCTION

Definite integrals are used to solve various problems. One of the usual applications is the calculation of the area of a plane region bounded by curves. This note presents different types of regions and gives the methods to calculate their areas.





AREA BETWEEN CURVES

If $f(x) \geq 0$ on $[a, b]$, then the area under curve $y = f(x)$ over $[a, b]$ is

$$\int_a^b f(x) dx.$$

If $f(x) \leq 0$ on $[a, b]$, then the area under curve $y = f(x)$ over $[a, b]$ is

$$\left| \int_a^b f(x) dx \right|$$

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AREA BETWEEN CURVES

Example 1.

Find the area between the graph of the function $f(x) = 5x - x^2$ and the x -axis.

Solution

To determine the limits of integration (the points at which $f(x)$ cuts the x -axis), we set $f(x) = 0$, that is,

$5x - x^2 = 0$, so that $x(5 - x) = 0$, implying that $x = 0$ or $x = 5$.

Taking any point, say 2 in $[0, 5]$, we see that $f(x) = f(2) = 5(2) - 2^2 = 6 > 0$. Therefore, the area is given by

$$\begin{aligned} A &= \int_0^5 f(x) dx = \int_0^5 5x - x^2 dx = \left[\frac{5x^2}{2} - \frac{x^3}{3} \right]_0^5 \\ &= \left[\frac{5(5)^2}{2} - \frac{5^3}{3} \right] - \left[\frac{5(0)^2}{2} - \frac{0^3}{3} \right] = \frac{125}{6} = 20.83 \end{aligned}$$

Hence, the area is 20.83 square units.





AREA BETWEEN CURVES

Example 2.

Find the area bounded by the curve $f(x) = x^2 - 1$, the x -axis and the lines $x = 0$ and $x = 2$.

Solution

The points at which $f(x)$ cuts the x -axis is given by $x^2 - 1 = 0$, so that $x = -1$ or $x = 1$. Since $x = 1$ is between 0 and 2, then the required region cuts x -axis at $x = 1$. In $[0,1]$, $f(x) \leq 0$ and in $[1,2]$, $f(x) \geq 0$. Therefore, the area is given by

$$\begin{aligned} A &= \left| \int_0^1 f(x) dx \right| + \int_1^2 f(x) dx = \left| \int_0^1 (x^2 - 1) dx \right| + \int_1^2 (x^2 - 1) dx \\ &= \left| \left[\frac{x^3}{3} - x \right]_0^1 + \left[\frac{x^3}{3} - x \right]_1^2 \right| \\ &= \left(\left| \frac{1^3}{3} - 1 - \left(\frac{0^3}{3} - 0 \right) \right| \right) + \left(\left[\frac{2^3}{3} - 2 \right] - \left[\frac{1^3}{3} - 1 \right] \right) = \frac{2}{3} + \frac{4}{3} = 2 \end{aligned}$$

Hence, the area is 2 square units.





TASK

Find the area between the graph of the function $f(x) = 6x - x^2$ and the x -axis.





AREA BETWEEN CURVES

If $f(x)$ and $g(x)$ are continuous with $f(x) \geq g(x)$ on $[a, b]$, then the area of the region between the curves $y = f(x)$ and $y = g(x)$ from a to b is given by

$$A = \int_a^b (f(x) - g(x))dx$$

If $f(x)$ and $g(x)$ are enclosed without a specific interval given, we let $f(x) = g(x)$, such that $x = a$ and $x = b$ are the bounds of the integration.





AREA BETWEEN CURVES

Example 1

Find the area bounded by $y = 3 - x$ and $y = x^2 - 9$.

Solution

Setting $3 - x = x^2 - 9$, we have $(x + 4)(x - 3) = 0$, so that $x = -4$ and $x = 3$. Let $f(x) = 3 - x$ and $g(x) = x^2 - 9$. Taking any point, say $x = 2$ in $[-4, 3]$, we see that $f(2) = 1$ and $g(2) = -5$. Therefore, $f(x) \geq g(x)$. The area is given by

$$\begin{aligned} A &= \int_{-4}^3 (f(x) - g(x)) dx = \int_{-4}^3 [(3 - x) - (x^2 - 9)] dx \\ &= \left[12x - \frac{x^3}{3} - \frac{x^2}{2} \right]_{-4}^3 = \frac{45}{2} - \frac{(-104)}{3} = \frac{343}{6} = 57.17 \end{aligned}$$

Hence, the required area is 57.17 square units





AREA BETWEEN CURVES

Example 2

Find the area enclosed by the curves $f(x) = 2 - x$ and $g(x) = 4 - x^2$.

Solution

Setting $f(x) = g(x)$, we have $x = -1$ and $x = 2$. Taking $x = 1$, we see that, $g(x) \geq f(x)$. Therefore, the area is given by

$$\begin{aligned} A &= \int_{-1}^2 (g(x) - f(x)) dx = \int_{-1}^2 [(4 - x^2) - (2 - x)] dx \\ &= \left[2x - \frac{x^3}{3} + \frac{x^2}{2} \right]_{-1}^2 = \frac{10}{3} - \frac{(-7)}{6} = \frac{9}{2} = 4.5 \end{aligned}$$

Hence, the required area is 4.5 square units.





TASK

Find the area bounded by $y = x^2$ and $y = 2 - x^2$ for $0 \leq x \leq 2$.





AREA BETWEEN CURVES

If $f(x)$ and $g(x)$ are defined on the interval $[a, b]$ and intersect on the interior, $c \in (a, b)$, we find all the intersections of $f(x)$ and $g(x)$ by letting $f(x) = g(x)$ such that $x = a$, $x = c$ and $x = b$ are the bounds of the integration. In that case, we have

$$A_1 = \int_a^c (f(x) - g(x))dx,$$

where $f(x) \geq g(x)$, and

$$A_2 = \int_c^b (g(x) - f(x))dx,$$

where $f(x) \leq g(x)$, so that the area bounded by $f(x)$ and $g(x)$ is given by

$$A = A_1 + A_2.$$





AREA BETWEEN CURVES

Example 1

Find the area bounded by $y = \sin(x)$ and $y = \cos(x)$ for $0 \leq x \leq \frac{\pi}{2}$.

Solution

To obtain the intersection points, let $\sin(x) = \cos(x)$. Then $\frac{\sin(x)}{\cos(x)} = 1$ or $\tan(x) = 1$, implying that $x = \frac{\pi}{4}$. Now, we know that $\cos(x) \geq \sin(x)$ when $0 \leq x \leq \frac{\pi}{4}$ and $\sin(x) \geq \cos(x)$ when $\frac{\pi}{4} \leq x \leq \frac{\pi}{2}$. Therefore, we have

$$\begin{aligned} A_1 &= \int_0^{\pi/4} (\cos(x) - \sin(x)) dx = [\sin(x) + \cos(x)]_0^{\pi/4} \\ &= \left(\frac{1}{\sqrt{2}} + \frac{1}{\sqrt{2}} \right) - (0 + 1) = \frac{2}{\sqrt{2}} - 1. \end{aligned}$$





AREA BETWEEN CURVES

Similarly,

$$\begin{aligned} A_2 &= \int_{\pi/4}^{\pi/2} (\sin(x) - \cos(x)) dx = [-\cos(x) - \sin(x)]_{\pi/4}^{\pi/2} \\ &= (0 - 1) - \left(-\frac{1}{\sqrt{2}} - \frac{1}{\sqrt{2}}\right) = -1 + \frac{2}{\sqrt{2}}. \end{aligned}$$

Therefore, the area bounded by $y = \sin(x)$ and $y = \cos(x)$ for $0 \leq x \leq \frac{\pi}{2}$ is given by

$$A = A_1 + A_2 = \left(\frac{2}{\sqrt{2}} - 1\right) + \left(-1 + \frac{2}{\sqrt{2}}\right) = 2\sqrt{2} - 2$$

Hence, the required area is $(2\sqrt{2} - 2)$ square units.





TASK

Find the area bounded by $y = \cos(x)$ and $y = 2 - \cos(x)$ for $0 \leq x \leq 2\pi$.





SUMMARY

The area bounded by the region of a function $f(x)$ defined on $[a, b]$ is given by

$$\int_a^b f(x) dx.$$

The area bounded by $f(x)$ and $g(x)$ defined on $[a, b]$ is given by

$$A = \int_a^b (f(x) - g(x)) dx$$

If $f(x)$ and $g(x)$ intersect at $c \in (a, b)$, then

$A_1 = \int_a^c (f(x) - g(x)) dx$, and $A_2 = \int_c^b (g(x) - f(x)) dx$,

so that the area bounded by $f(x)$ and $g(x)$ is given by

$$A = A_1 + A_2.$$





FURTHER READING

"Calculus" by James Stewart.

"Calculus: Early Transcendentals" by Howard Anton, Irl Bivens, and Stephen Davis.

"Calculus Know-It-ALL" by Stan Gibilisco.

"Theory and Problems of ADVANCED CALCULUS" by Robert Wrede, and Murray R. Spiegel.

"Advanced Engineering Mathematics" by H. K. Dass.

"Engineering Mathematics" by K. A. Stroud.





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THANK YOU